Self-supervised Learning
Autoencoders

Hermann Blum

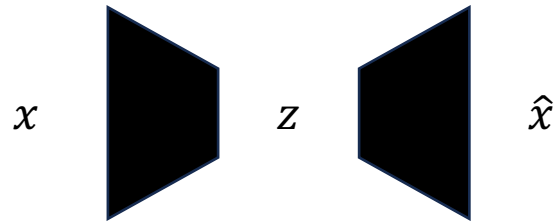
UNIVERSITÄT BONN

LAMARR INSTITUTE FOR MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

Robot Perception and Learning Lab
rpl.uni-bonn.de

General Concept

- Compress the information in x into a latent represention z, from which a reconstruction of x can be recovered
  - removes any redundant information
  - reduces dimensionality
  - may discover abstract patterns



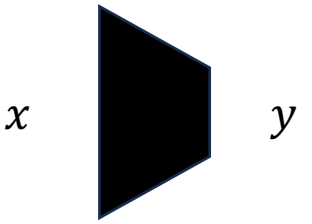
Hyperparameters

- dimensionality of z
  usually we want this much smaller than x
- capacity of the encoder/decoder
  if too small: the loss will not converge
  if too large: the autoencoder can just store the dataset
- regularization
  different techniques to shape the latent space z

Information Bottleneck

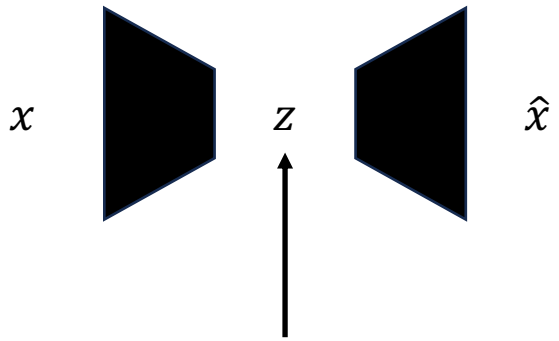
In neural networks in general, for a hidden layer t, compare $I(X,T)$ with $I(T,Y)$

➔we want a tradeoff keeping all task-relevant information of x while ignoring all irrelevant information



$I(X,T)$ depends on capacity, but also architecture

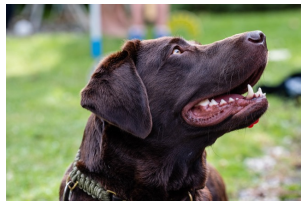
Information Bottleneck



Autoencoders are designed such that z is the information bottleneck.

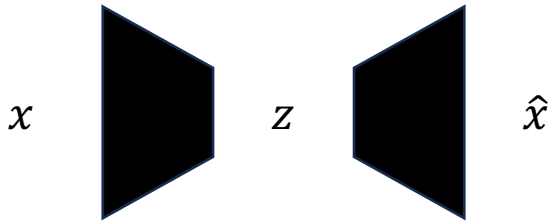
➔ We usually want a z that contains all relevant information of x that is explainable from the dataset X and not noise.

What will the AE learn?



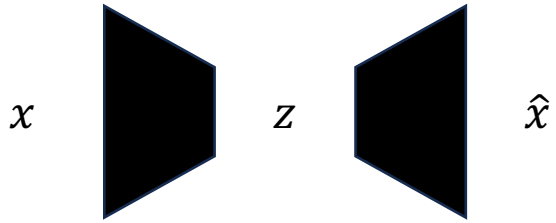






➔even if z is one-dimensional, with enough capacity in encoder and decoder the AE can learn a codebook into our limited dataset

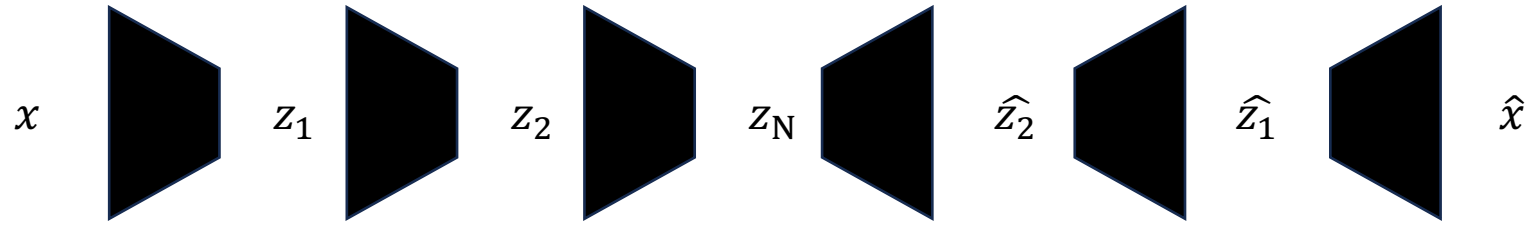
How can we train an Autoencoder?



simple reconstruction loss $\mathcal{L} = \|x - \hat{x}\|_2$

used to be very hard to train for more than 3 layers because of vanishing gradients

Stacked Autoencoder

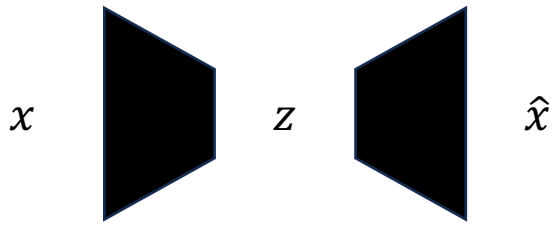


outer most layer is trained on $\mathcal{L} = \|x - \hat{x}\|_2$

inner layers trained on $\mathcal{L} = \|z - \hat{z}\|_2$

ever layer can be shallow, but they can be „stacked“ to form a deep autoencoder

How can we train an Autoencoder?

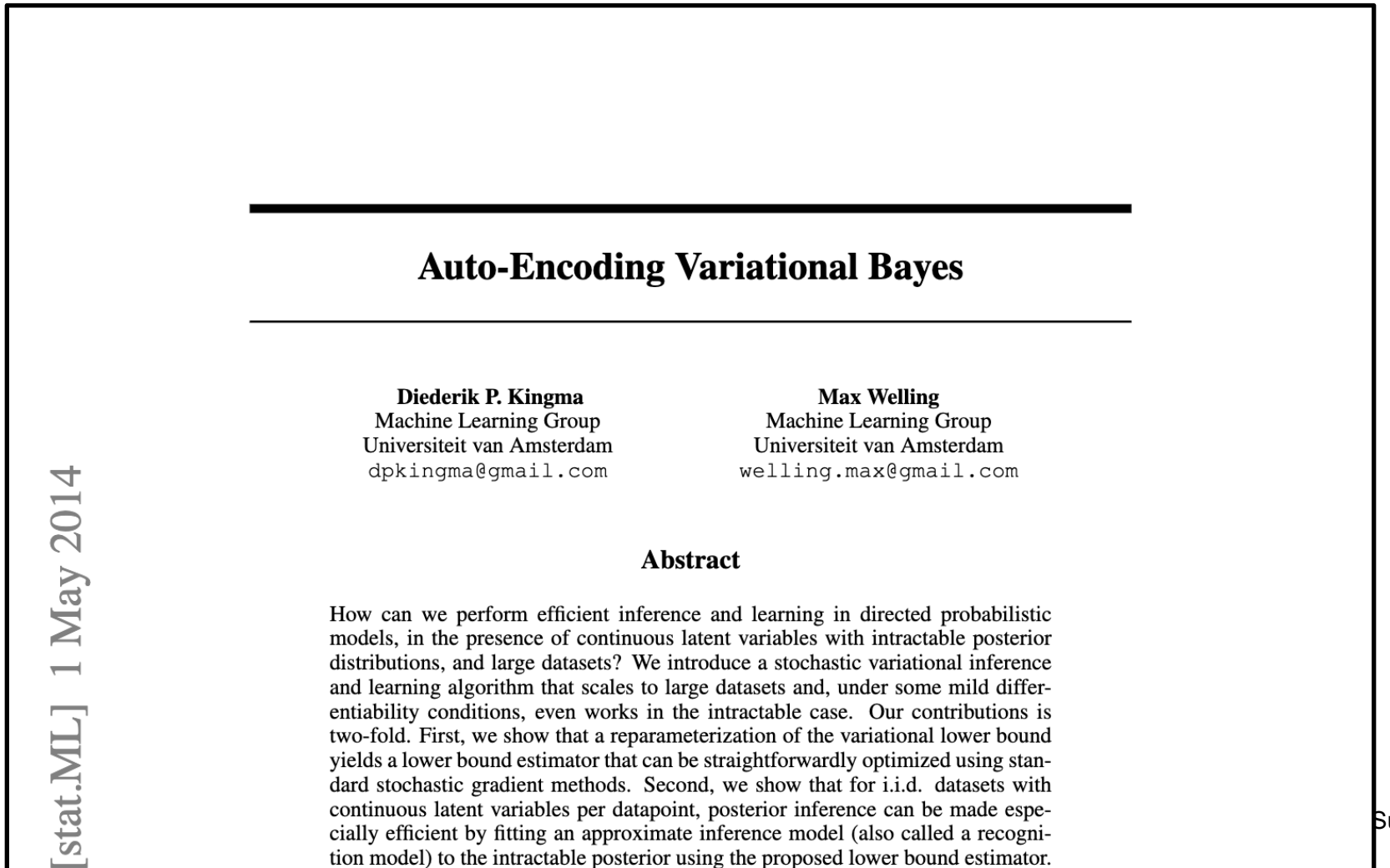


simple reconstruction loss $\mathcal{L} = \|x - \hat{x}\|_2$

used to be very hard to train for more than 3 layers because of vanishing gradients

usually not a problem anymore with BatchNorm

Variational Autoencoders



arXiv:1312.6114v10 [stat.ML] 1 May 2014

Auto-Encoding Variational Bayes

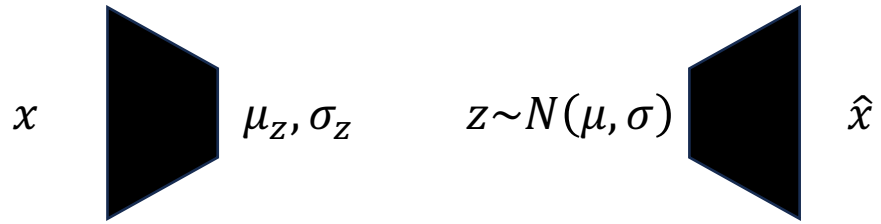
Diederik P. Kingma
Machine Learning Group
Universiteit van Amsterdam
dpkingma@gmail.com

Max Welling
Machine Learning Group
Universiteit van Amsterdam
welling.max@gmail.com

Abstract

How can we perform efficient inference and learning in directed probabilistic models, in the presence of continuous latent variables with intractable posterior distributions, and large datasets? We introduce a stochastic variational inference and learning algorithm that scales to large datasets and, under some mild differentiability conditions, even works in the intractable case. Our contributions is two-fold. First, we show that a reparameterization of the variational lower bound yields a lower bound estimator that can be straightforwardly optimized using standard stochastic gradient methods. Second, we show that for i.i.d. datasets with continuous latent variables per datapoint, posterior inference can be made especially efficient by fitting an approximate inference model (also called a recognition model) to the intractable posterior using the proposed lower bound estimator.

Variational Autoencoders



treat z and x as random variables

assume a prior on z: $\mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$

learn distributions $\mathrm{q}_\theta(\mathbf{z} \mid \mathbf{x})$ and $\mathrm{p}_\phi(\mathbf{x} \mid \mathbf{z})$

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}^{(i)}) = -D_{KL}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})||p_{\boldsymbol{\theta}}(\mathbf{z})) + \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}|\mathbf{z})\right]$$

Variational Autoencoders

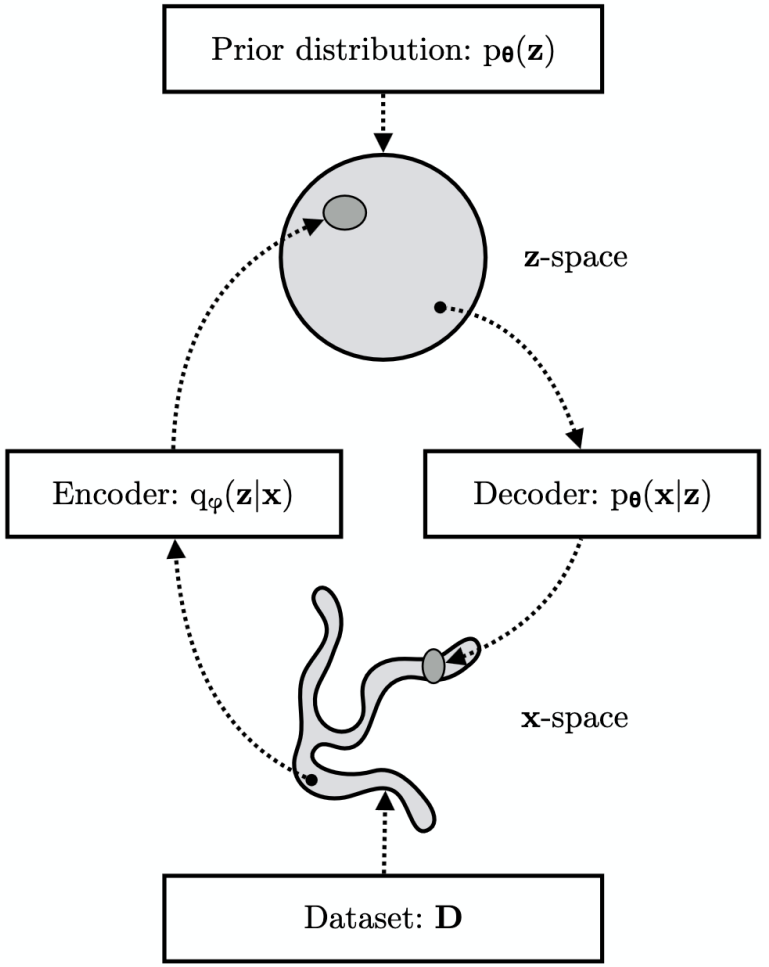


Figure from Kingma & Welling 2019

What is ELBO?

ELBO = Evidence Lower Bound

a lower bound on the log-likelihood

We obtain the lower bound using Jensen's Inequality: for a convex function h of a random variable X,

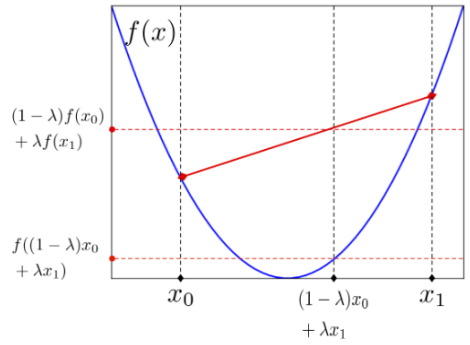
$$\mathbb{E}[h(X)] \geq h(\mathbb{E}[X])$$

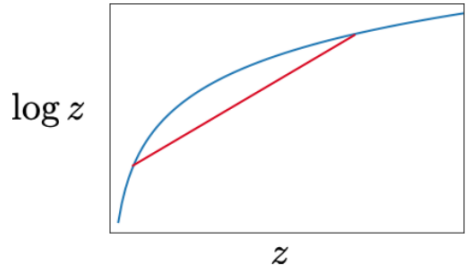
Therefore, if h is concave (i.e. $-h$ is convex),

$$\mathbb{E}[h(X)] \leq h(\mathbb{E}[X])$$

The function $\log z$ is concave. Therefore,

$$\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$$



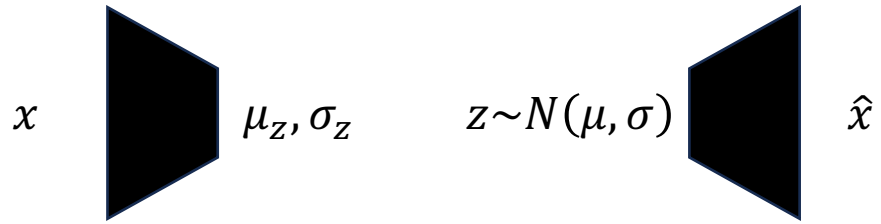


What is ELBO?

$$
\begin{aligned}
\log p(\mathbf{x}) &= \log \int p(\mathbf{z})\, p(\mathbf{x}|\mathbf{z})\, \mathrm{d}\mathbf{z} \\
&= \log \int q(\mathbf{z})\, \frac{p(\mathbf{z})}{q(\mathbf{z})} p(\mathbf{x}|\mathbf{z})\, \mathrm{d}\mathbf{z} \\
&\geq \int q(\mathbf{z}) \log \left[\frac{p(\mathbf{z})}{q(\mathbf{z})}\, p(\mathbf{x}|\mathbf{z}) \right] \mathrm{d}\mathbf{z} \qquad \text{(Jensen's Inequality)} \\
&= \mathbb{E}_q \left[\log \frac{p(\mathbf{z})}{q(\mathbf{z})} \right] + \mathbb{E}_q \left[\log p(\mathbf{x}|\mathbf{z}) \right]
\end{aligned}
$$

slide by Grosse & Ba, U Toronto

Variational Autoencoders



treat z and x as random variables

assume a prior on z: $\mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$

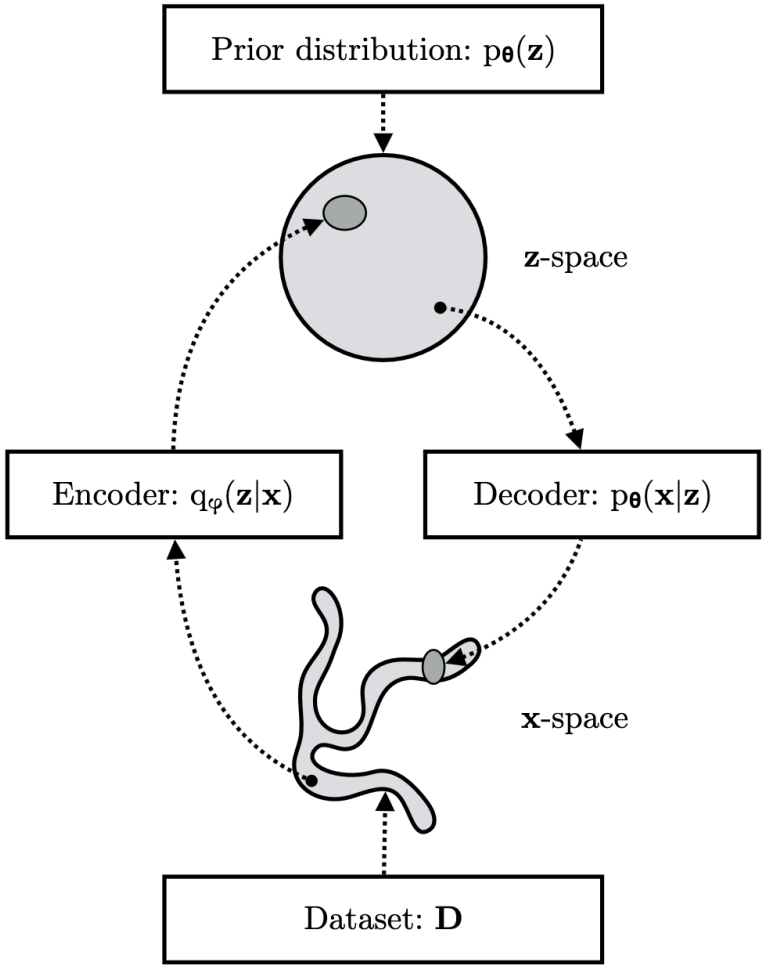
learn distributions $\mathrm{q}_\theta(\mathbf{z} \mid \mathbf{x})$ and $\mathrm{p}_\phi(\mathbf{x} \mid \mathbf{z})$

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}^{(i)}) = -D_{KL}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})||p_{\boldsymbol{\theta}}(\mathbf{z})) + \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}|\mathbf{z})\right]$$

Decoder Objective

- The first term we'll look at is $\mathbb{E}_q\left[\log p(\mathbf{x}|\mathbf{z})\right]$
- Since we assumed a Gaussian observation model,

$$\begin{aligned}\log p(\mathbf{x}|\mathbf{z}) &= \log \mathcal{N}(\mathbf{x}; G_{\boldsymbol{\theta}}(\mathbf{z}), \eta \mathbf{I}) \\ &= \log\left[\frac{1}{(2\pi\eta)^{D/2}} \exp\left(-\frac{1}{2\eta}\|\mathbf{x} - G_{\boldsymbol{\theta}}(\mathbf{z})\|^2\right)\right] \\ &= -\frac{1}{2\eta}\|\mathbf{x} - G_{\boldsymbol{\theta}}(\mathbf{z})\|^2 + \text{const}\end{aligned}$$



Reparametrization Trick

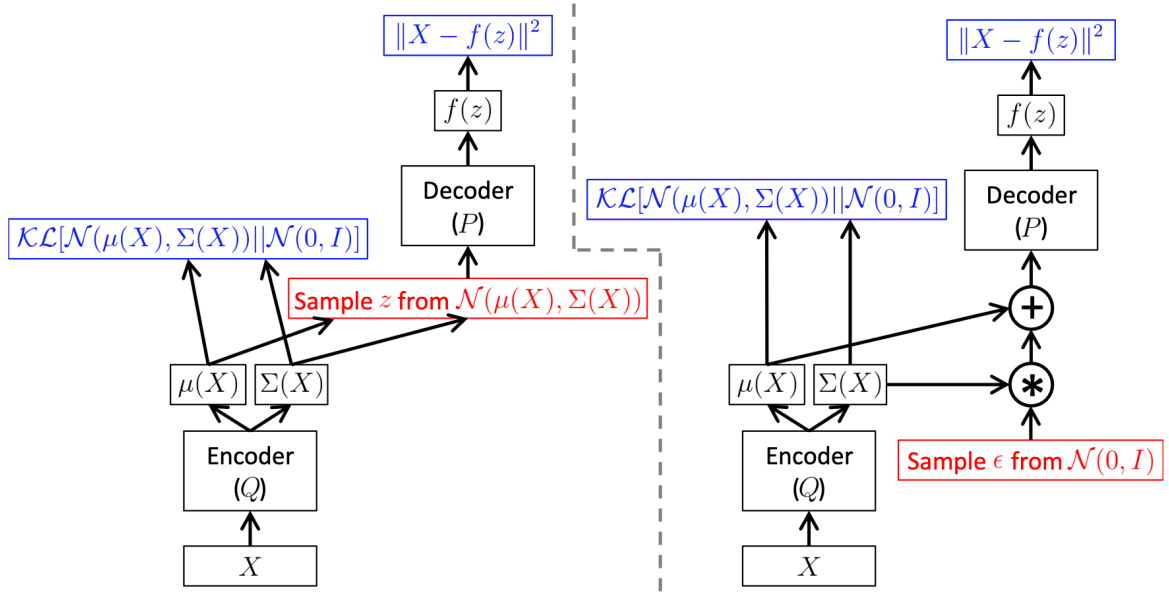


Figure 4: A training-time variational autoencoder implemented as a feed-forward neural network, where $P(X|z)$ is Gaussian. Left is without the "reparameterization trick", and right is with it. Red shows sampling operations that are non-differentiable. Blue shows loss layers. The feedforward behavior of these networks is identical, but backpropagation can be applied only to the right network.

Figure from Doersch (2016)

Variational Autoencoders

the true ELBO objective

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}^{(i)}) = -D_{KL}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})||p_{\boldsymbol{\theta}}(\mathbf{z})) + \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}^{(i)})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}|\mathbf{z})\right]$$

gets approximated as

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{x}^{(i)}) \simeq \frac{1}{2}\sum_{j=1}^{J}\left(1 + \log((\sigma_j^{(i)})^2) - (\mu_j^{(i)})^2 - (\sigma_j^{(i)})^2\right) + \frac{1}{L}\sum_{l=1}^{L}\log p_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}|\mathbf{z}^{(i,l)})$$

where $\mathbf{z}^{(i,l)} = \boldsymbol{\mu}^{(i)} + \boldsymbol{\sigma}^{(i)} \odot \boldsymbol{\epsilon}^{(l)}$ and $\boldsymbol{\epsilon}^{(l)} \sim \mathcal{N}(0, \mathbf{I})$

with

where $\ln p_\theta(x|z)$ is implemented as $-\frac{1}{2}\|x - D_\theta(z)\|_2^2$,

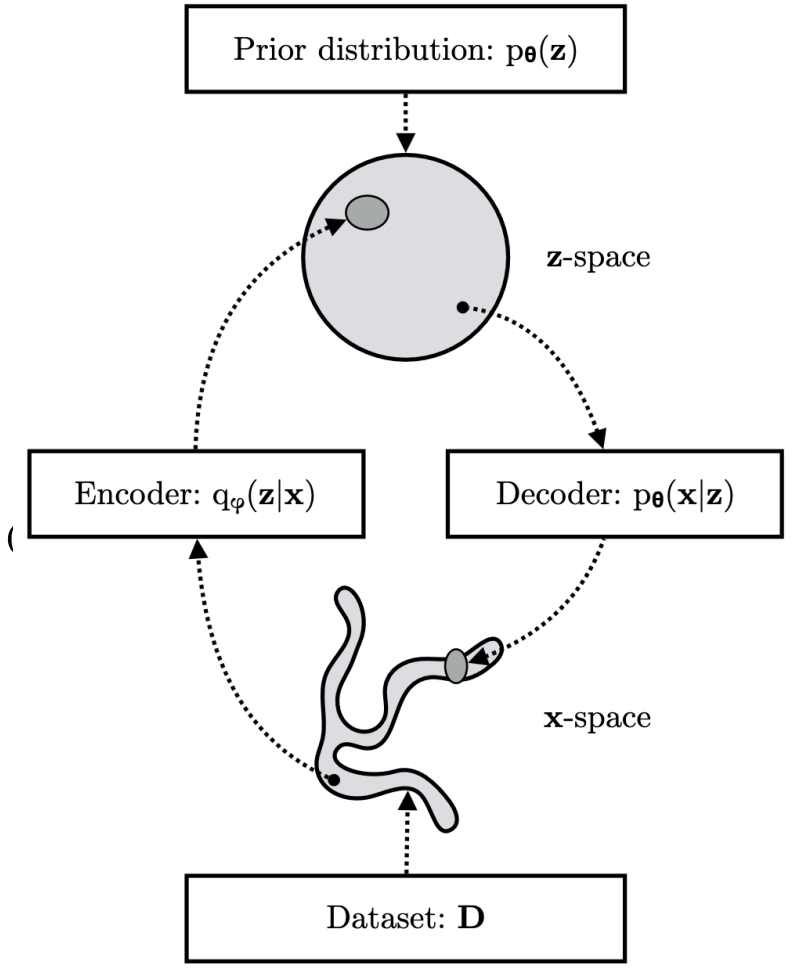


Figure from Kingma & Welling 2019



Figure from Kingma & Welling 2014

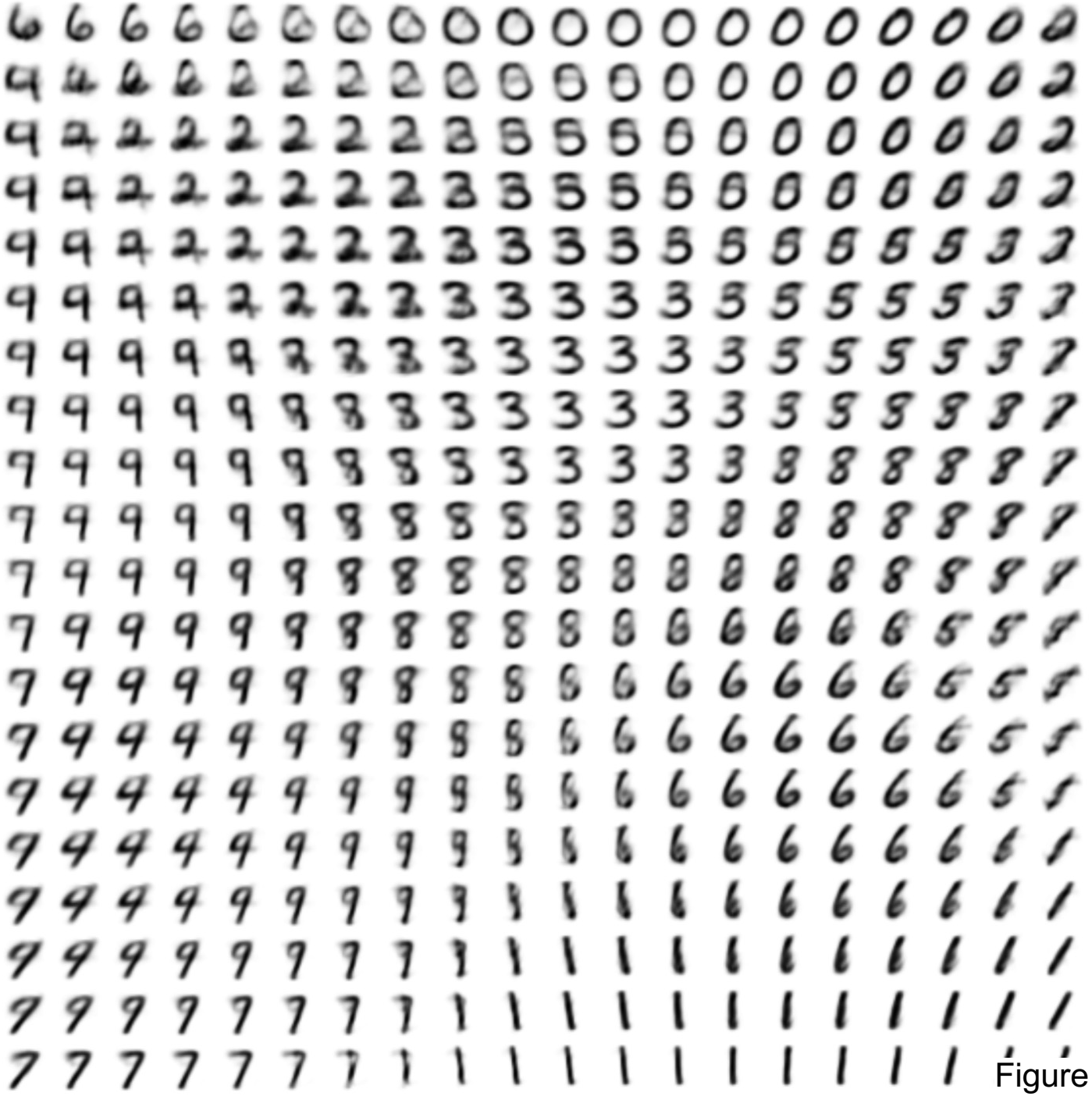


Figure from Kingma & Welling 2014

VAE is a Generative Model



Samples from a Data Distribution

Train



Neural Network



Sample



slide from https://cvpr2022-tutorial-diffusion-models.github.io/

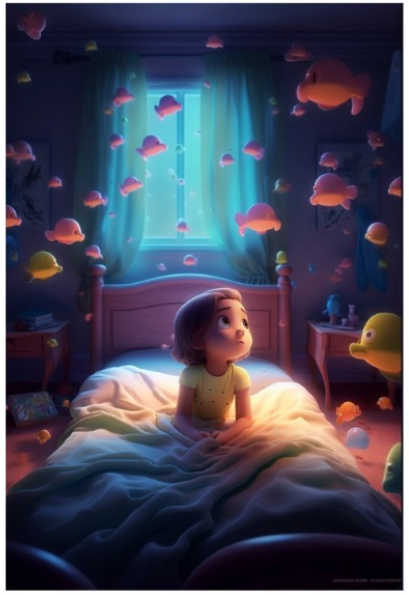
AI Art















Midjourney v5

Slide by Ruiqi Gao @Stanford cs231n

Masked Autoencoder (MAE)

This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Masked Autoencoders Are Scalable Vision Learners

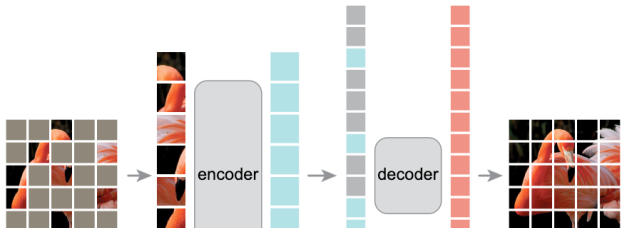
Kaiming He*,† Xinlei Chen* Saining Xie Yanghao Li Piotr Dollár Ross Girshick

*equal technical contribution †project lead

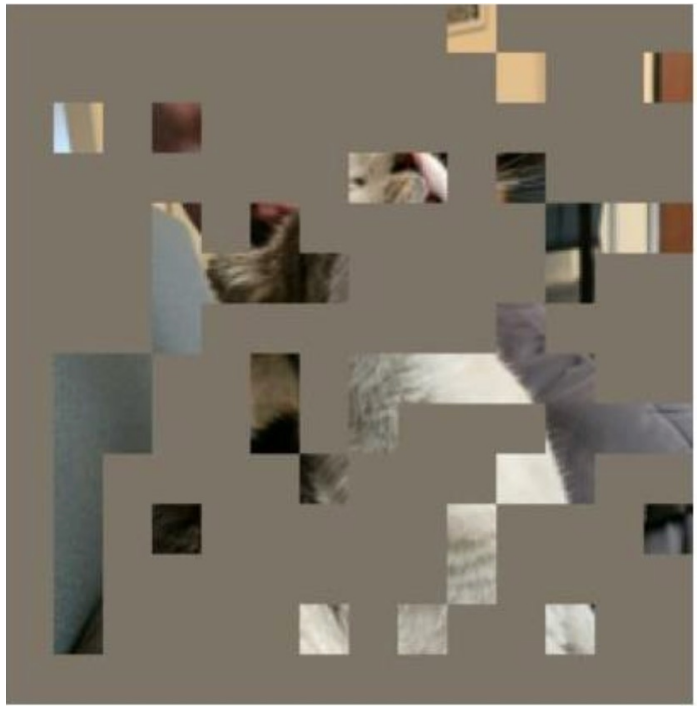
Facebook AI Research (FAIR)

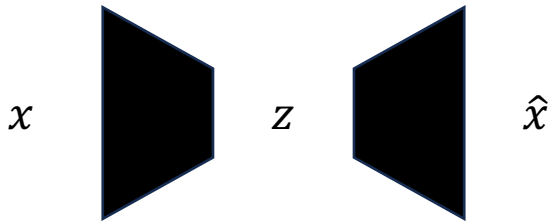
Abstract

This paper shows that masked autoencoders (MAE) are scalable self-supervised learners for computer vision. Our MAE approach is simple: we mask random patches of the input image and reconstruct the missing pixels. It is based on two core designs. First, we develop an asymmetric



Masked Autoencoder (MAE)







He et al., 2021

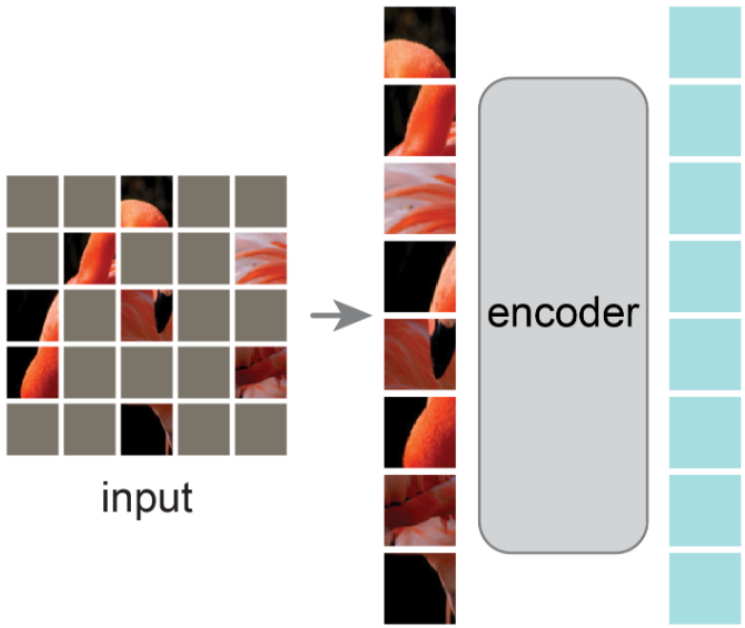
How MAE Works?



Random masking

Slide by Xinlei Chen, ECCV‘22 MAE Tutorial

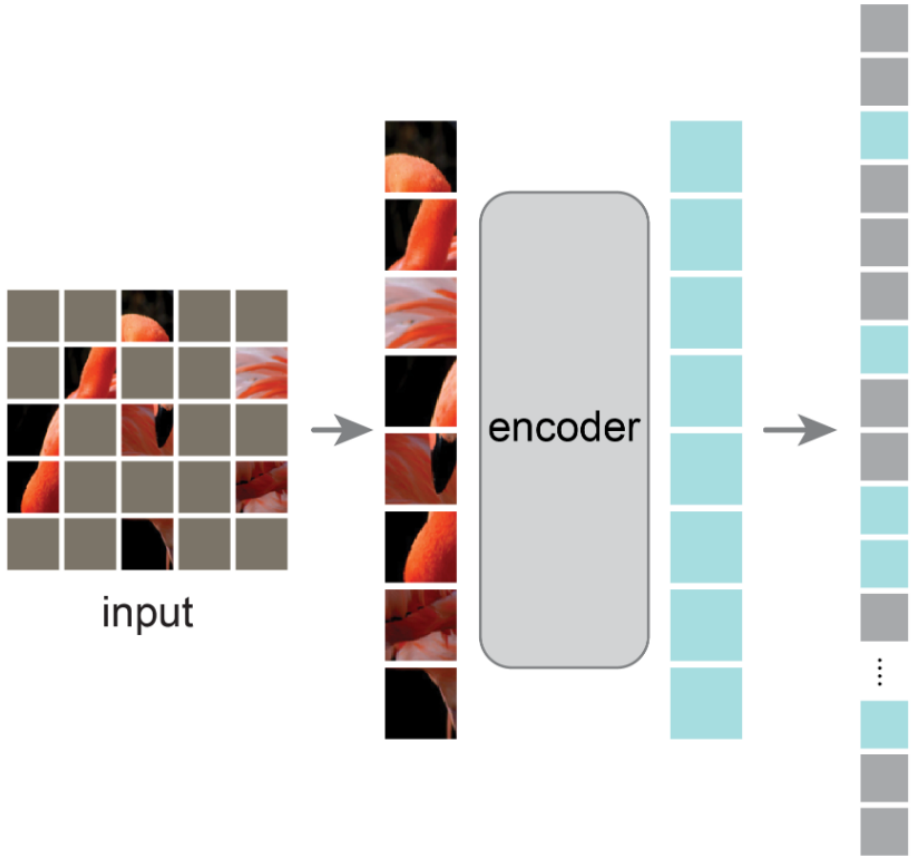
How MAE Works?



Encode visible patches

Slide by Xinlei Chen, ECCV'22 MAE Tutorial

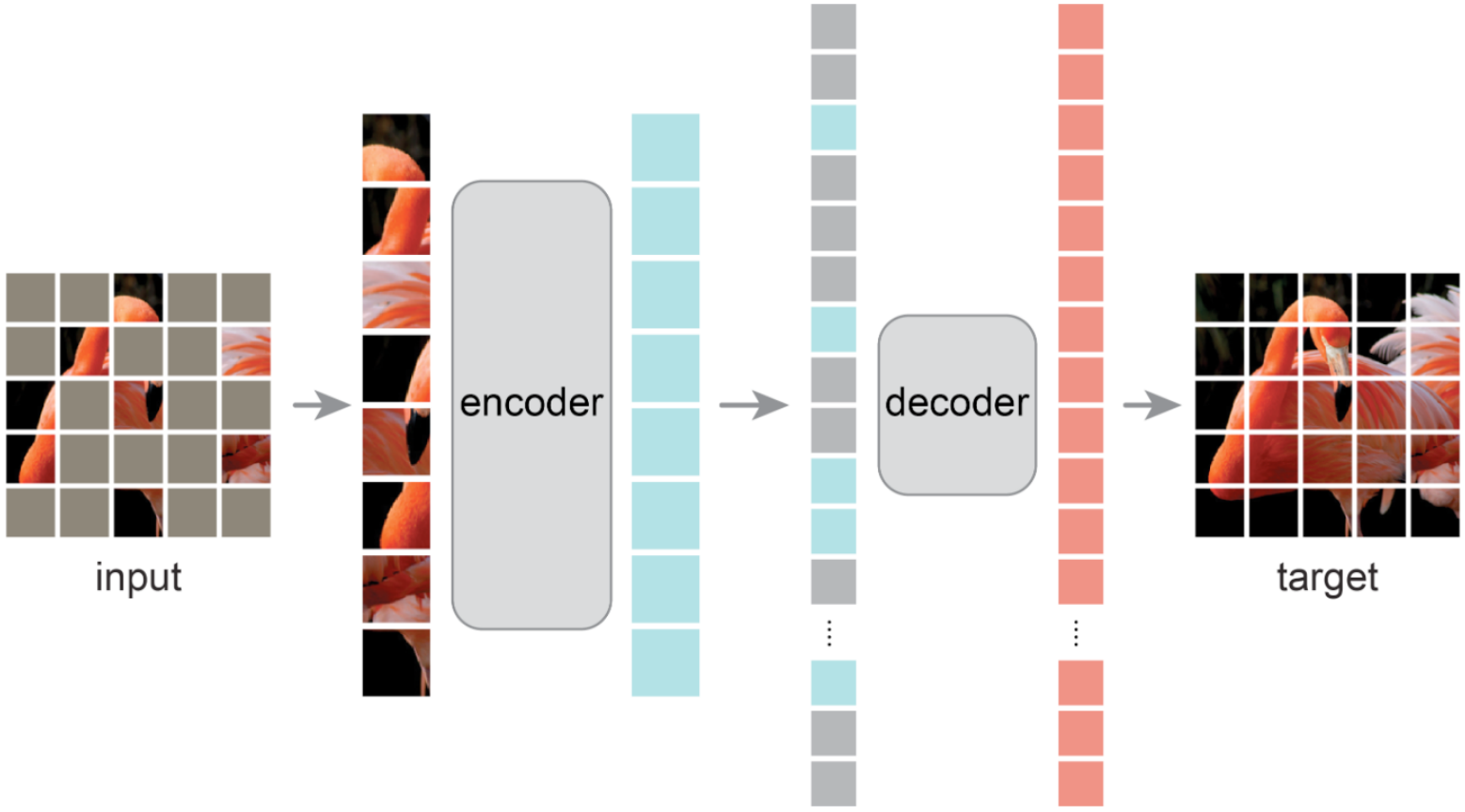
How MAE Works?



Add mask tokens

Slide by Xinlei Chen, ECCV'22 MAE Tutorial

How MAE Works?



Slide by Xinlei Chen, ECCV'22 MAE Tutorial

Masking Methos

- Similar to the original ViT,
  divide the input into non-overlapping patches.
- Uniformly sample a very large proportion (75%) of these patches and mask them
- Masking a high ratio makes predicting the task challenging and meaningful.
- Also, not using mask tokens and picking a high sampling ratio (masking most of the image, e.g., 75%, and sampling a small visible portion, e.g., 25%, to feed into the encoder) enables the encoder to be very large.

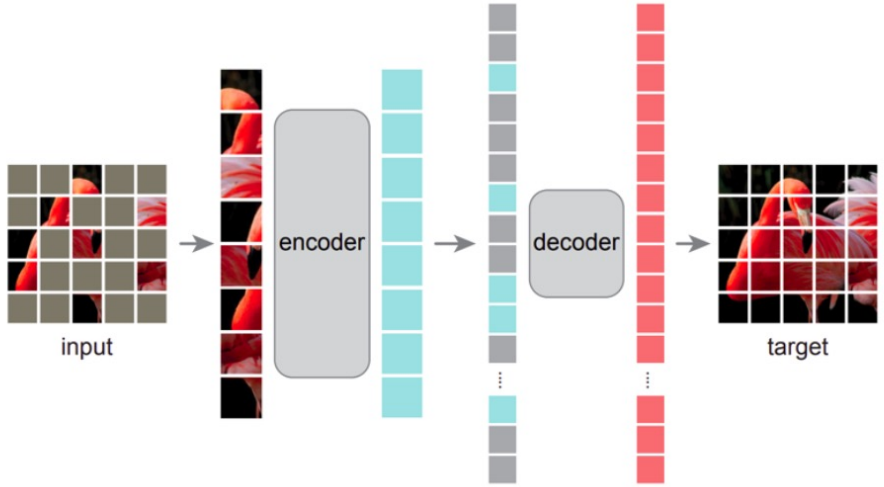


Figure 6. MAE architecture from the paper

He et al., 2021 Masked Autoencoders Are Scalable Vision Learners

MAE Encoder

- The encoder only operates on unmasked patches (25%)
- Embeds the patches by linear projection and add positional embeddings
- Uses transformer blocks
- Since the input patches are a small part of the input, the encoder is chosen to be very large. (encoder has over 9 times computations per token vs decoder)

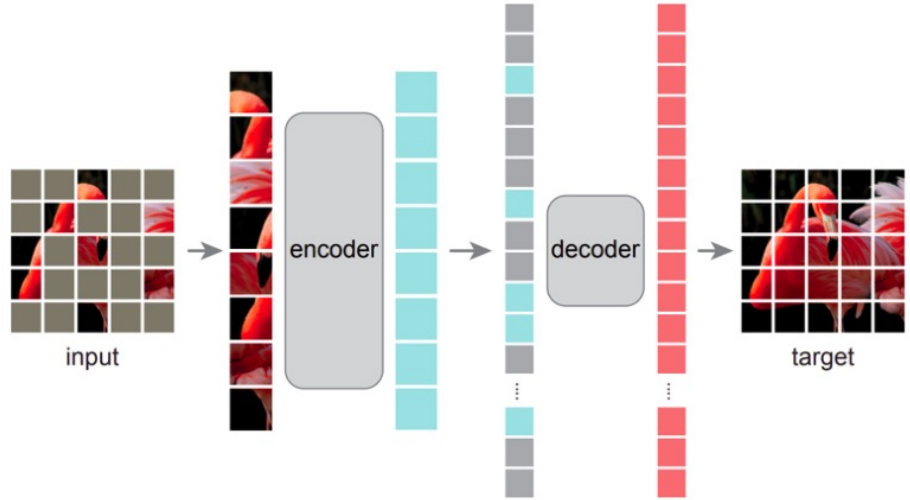


Figure 6. MAE architecture from the paper

He et al., 2021 Masked Autoencoders Are Scalable Vision Learners

Slide by Fei-Fei, Adeli, Durante, Johnson, Xiang, Stanford cs231n

MAE Decoder

- Merges the encoder outputs with the shared mask tokens in previously masked places, adding positional encodings to them.
- Uses transformer blocks, followed by a linear projection for finalizing pixel reconstruction.
- Is solely responsible for reconstruction, meaning it is not used post-training. Hence, it is independent of the encoder design, making it flexible (unlike traditional AEs or UNet). This is an asymmetrical autoencoder design.

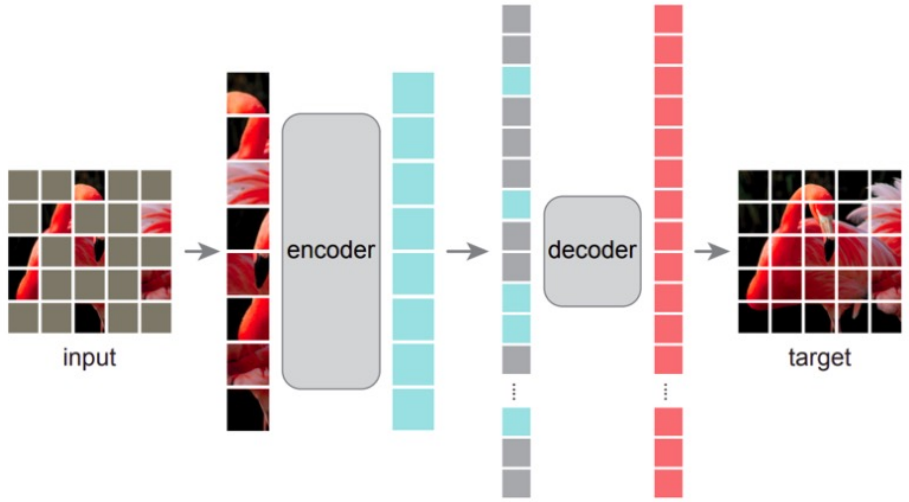


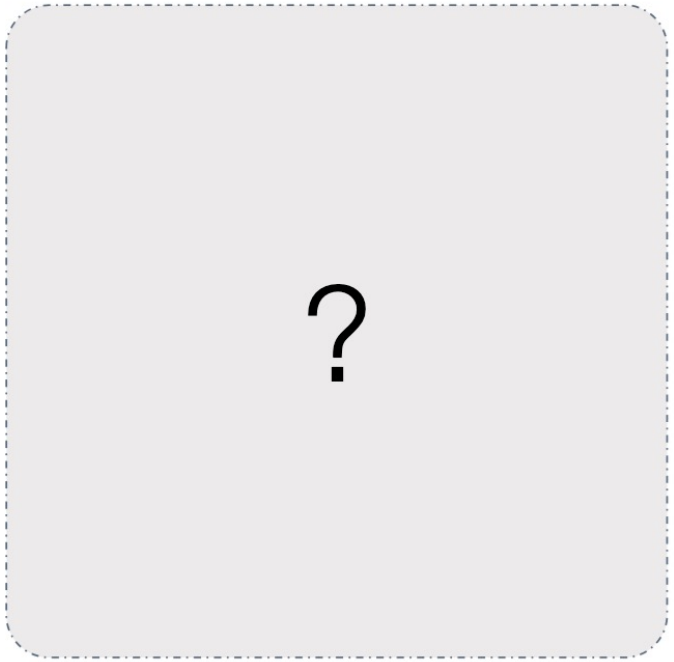
Figure 6. MAE architecture from the paper

He et al., 2021 Masked Autoencoders Are Scalable Vision Learners

MAE Examples



Masked input: 80%

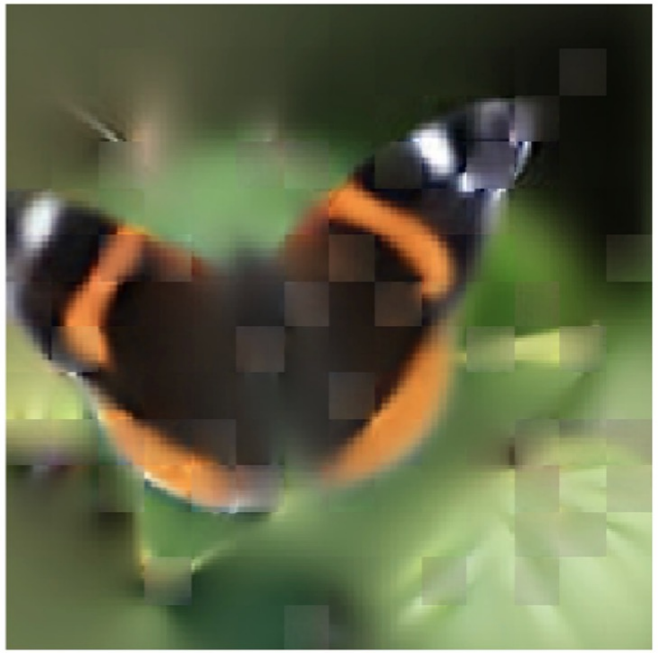


You guess?

MAE Examples

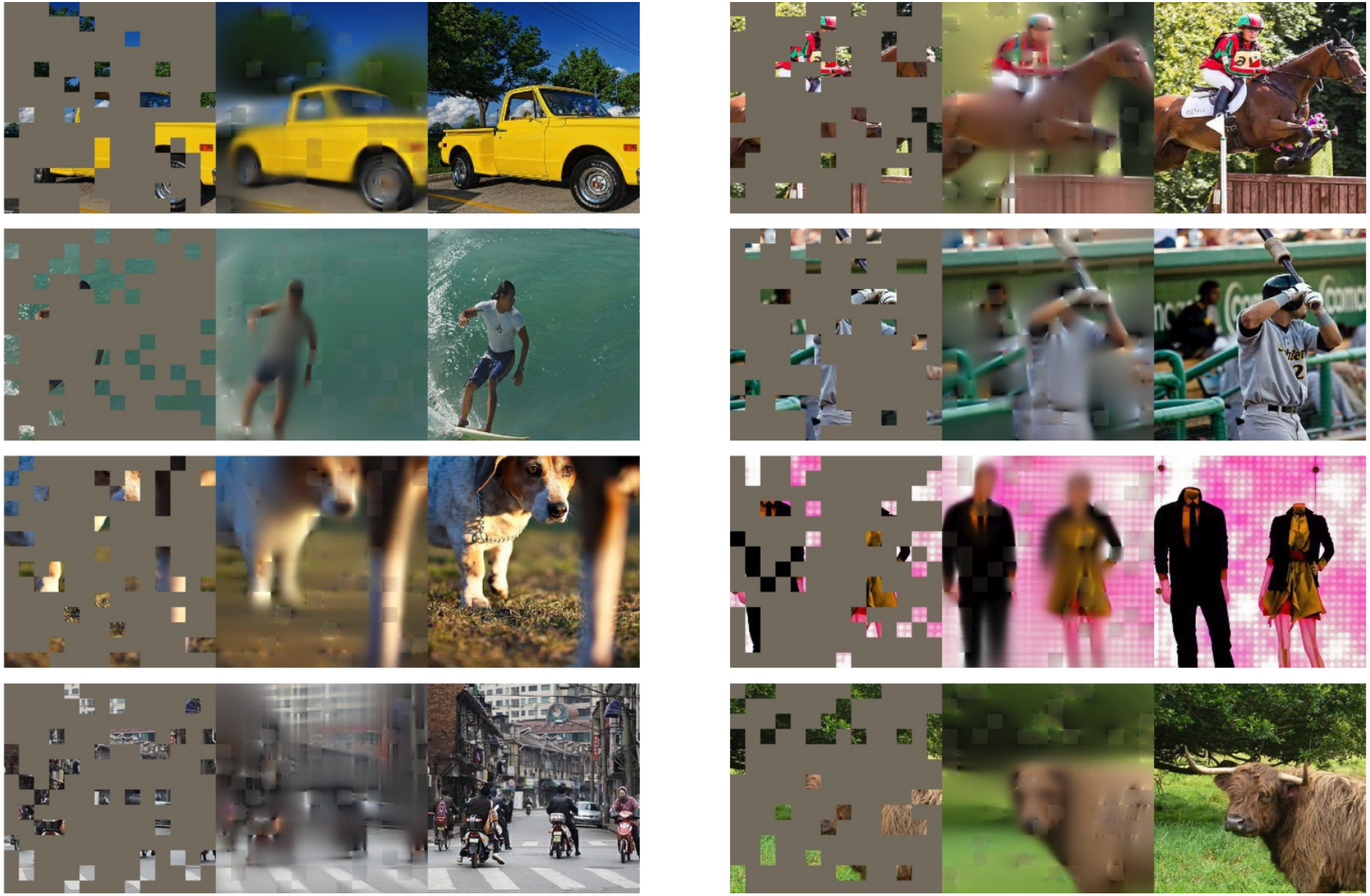


Masked input: 80%



MAE's guess

Slide by Xinlei Chen, ECCV'22 MAE Tutorial



MAE Examples



original



75% mask



85% mask

95% mask



MAE Results

Train ViT-L from scratch on ImageNet: 82.5 % Acc

Pre-train with MAE, then finetune: 84.9 % Acc

the same pretraining can be useful for other tasks:

		AP^{box}		AP^{mask}	
method	pre-train data	ViT-B	ViT-L	ViT-B	ViT-L
supervised	IN1K w/ labels	47.9	49.3	42.9	43.9
MoCo v3	IN1K	47.9	49.3	42.7	44.0
BEiT	IN1K+DALLE	49.8	**53.3**	44.4	47.1
MAE	IN1K	**50.3**	**53.3**	**44.9**	**47.2**

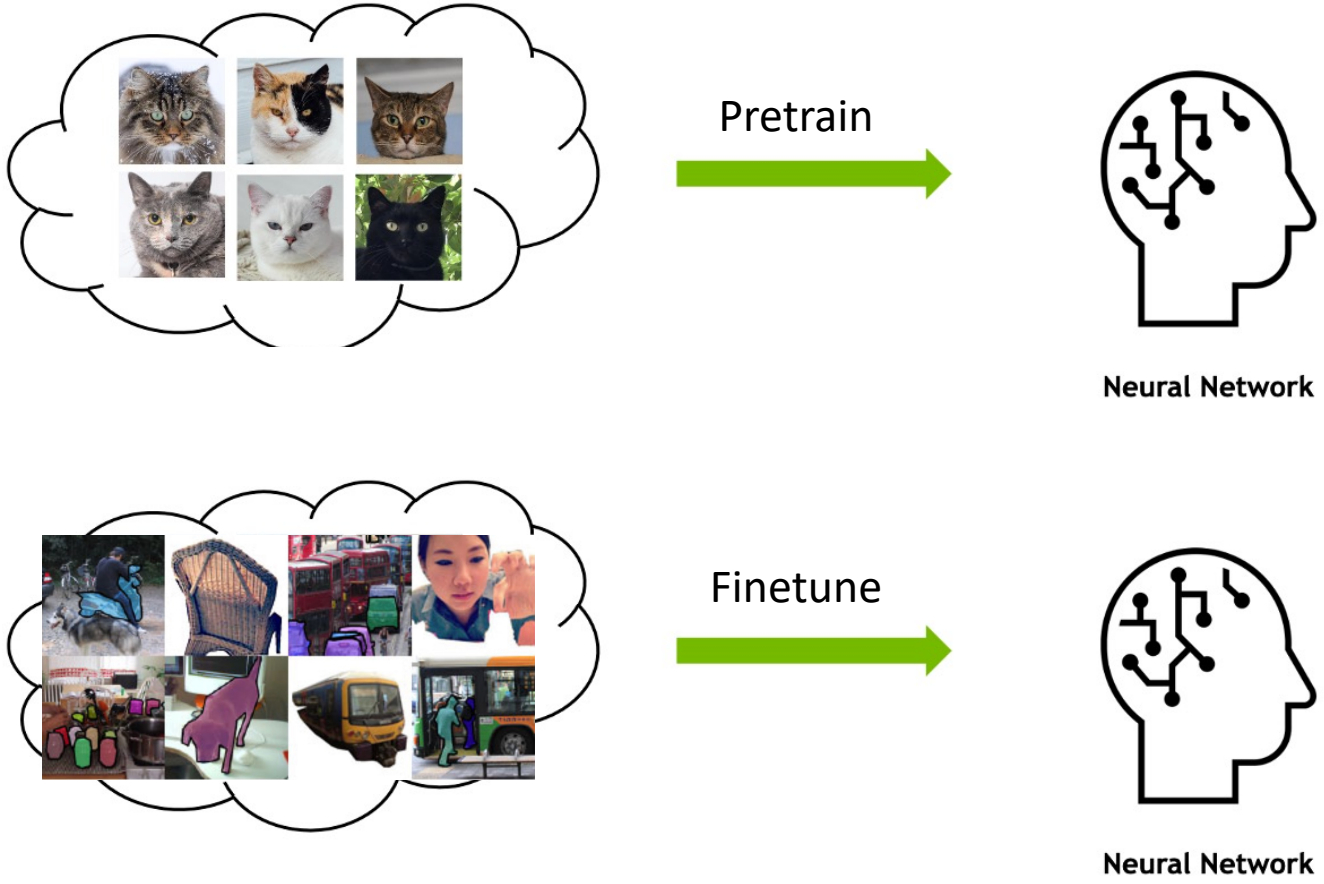
Table 4. **COCO object detection and segmentation** using a ViT Mask R-CNN baseline. All entries are based on our implementation. Self-supervised entries use IN1K data *without* labels. Mask AP follows a similar trend as box AP.

method	pre-train data	ViT-B	ViT-L
supervised	IN1K w/ labels	47.4	49.9
MoCo v3	IN1K	47.3	49.1
BEiT	IN1K+DALLE	47.1	53.3
MAE	IN1K	**48.1**	**53.6**

Table 5. **ADE20K semantic segmentation** (mIoU) using UperNet. BEiT results are reproduced using the official code. Other entries are based on our implementation. Self-supervised entries use IN1K data *without* labels.

He et al., 2021

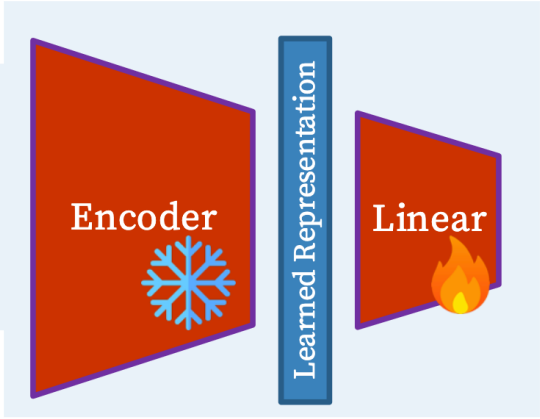
MAE is a Representation Learner



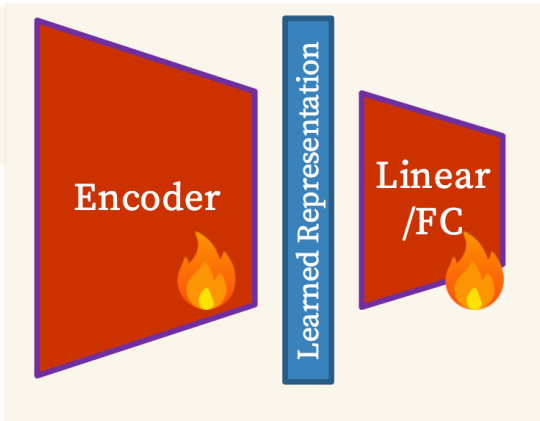
MAE is a Representation Learner

Linear Probing vs Full Fine-tuning

- In linear probing, the pre-trained model is fixed, and only one linear layer is added at the end, to predict the labels (or produce the output). This method is used to assess the quality of representations from a pre-trained feature extraction model.



- In fine-tuning, pre-trained model is further trained (not fixed), and one or more layers, possibly with non-linearities are added.



- linear probing: provides a measure of representation quality of a pre-training in restricted conditions
- fine-tuning: exploits models near-true potential to adopt for new tasks

He et al., 2021 Masked Autoencoders

MAE is a Representation Learner

